ModelIt!

Student Activity Pack

Why Do Fireworks Explode in Different Colors?

Modeling Electron Energy Transitions and Emission Spectra

Grade 10L1 | NGSS HS-PS1-1, HS-PS1-2 | Patterns

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Why do you think fireworks come in so many different colors — where does the color come from?**

2. **What do you think happens to electrons inside an atom when it is heated to extreme temperatures?**

3. **Draw a diagram of what you think is happening inside an atom when it gives off colored light.**

4. **If two different metals are burned and produce different colors, what does that tell you about their atoms?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Element Type | Electron Energy Level | Oxidizer Strength | Combustion Temperature | Color Wavelength

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? **Explain ONE relationship in your model. What happens when one component changes?**

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Strontium Red

Element: Strontium chloride | Oxidizer: Moderate

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Copper Blue

Element: Copper chloride | Oxidizer: High

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Temperature Override

Element: Strontium | Oxidizer: Maximum

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Atmospheric Oxygen — The concentration of oxygen available for combustion, which affects the completeness of the reaction and whether metal atoms fully oxidize or remain in colored ionic states

Shell Burst Pattern — The physical design of the firework shell that determines how metal salts are distributed during detonation, affecting the spatial distribution and mixing of colors

Spectral Purity — A measure of how clean and saturated the emitted color is, affected by contamination from other elements, blackbody radiation, and incomplete combustion

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can burning the same gunpowder produce red, blue, green, and gold — and what does that tell us about the hidden structure of atoms? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that color is a property of atomic structure rather than the chemical reaction itself?**

2. **Why does increasing combustion temperature beyond a certain point actually reduce color quality?**

3. **How could astronomers use the same principles as fireworks to identify elements in stars billions of miles away?**

4. **What would happen to the firework's color if you mixed two different metal salts in the same shell?**

5. **What are the limitations of modeling quantum electron transitions with only five components?**

8 STEM Challenge: Design a Custom Fireworks Color Sequence

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YOUR ENGINEERING MISSION

Design a firework shell that produces a specific three-color sequence by selecting the right metal salts and optimizing oxidizer ratios for maximum color purity.

THE SCENARIO

A fireworks company has been hired to create a custom show for a city's 250th anniversary. The city's colors are deep red, emerald green, and royal blue. Your team must design shells that produce these exact colors in sequence, with maximum brightness and color purity. The company needs your spectral analysis and chemical formulations.

Engineering Design Thinking

1. WHAT SPECIFIC METAL COMPOUNDS WILL YOU USE FOR EACH COLOR AND WHAT IS THE SCIENTIFIC BASIS FOR YOUR CHOICE?

2. HOW WILL YOU PREVENT COLORS FROM BLENDING WHEN MULTIPLE SHELLS BURST IN SEQUENCE?

3. WHAT OXIDIZER CONCENTRATIONS WILL YOU USE AND HOW DID YOUR MODEL DATA INFORM THIS DECISION?

4. HOW WOULD YOU TEST YOUR FORMULATIONS BEFORE A LIVE SHOW?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Pyrotechnic Chemists design firework formulations by applying quantum mechanics and combustion chemistry. They work for fireworks manufacturers, special effects companies, and military ordnance programs, earning $50,000–$95,000/year. Spectroscopists who analyze light from atoms earn $65,000–$120,000/year in research and industry.